

# Discovery Executive Summary

Project: discovery-14-aug · Generated: 14/08/2026, 11:56:19

EXECUTIVE SUMMARY

This report consolidates the overall ratings, key findings, and recommended actions from the 1 discovery analysis run across this codebase (frontend and backend). Each section below reproduces that analysis's executive view; full evidence and diagrams live in the individual reports.

## Portfolio Overview

| # | Analysis                           | Overall Rating |
|---|------------------------------------|----------------|
| 1 | Code Quality & Complexity Analysis | —              |

## 1. Code Quality & Complexity Analysis

EXECUTIVE SUMMARY

The Klearcom platform codebase is a small monorepo (~3,008 LOC) with a Laravel backend, a React/TypeScript frontend, and a Node.js development API. Overall code quality is reasonable for an early-stage project, but one structural issue rises to High Risk: the `ConnectPage` React component is a 214-line single function that mixes form state, three TanStack Query subscriptions, mutation logic, event handlers, and deeply nested conditional JSX — exceeding the 200 LOC function threshold. Cyclomatic complexity is Moderate, peaking at ~12 branches in the same component. Duplicate code sits at ~5% overall, driven by identical `buildTree` implementations in three locations and a reachability-calculation block copied across `ConnectController`, `RealTimeTestService`, and the dev-API. Git churn and ownership metrics are healthy (3 total commits, single author), so stability risks are minimal. One additional hotspot — a resource-leak anti-pattern in `LegacyMonitorPoller.jsx` — was identified. Analysis covered both backend (26 PHP files) and frontend (15 TS/TSX/JSX files) layers in full.

### 2.1 Benchmark Ratings Summary

| #  | Hotspot                    | Primary KPI                     | <span class="rating rating-good">Good | <span class="rating rating-moderate">Moderate | <span class="rating rating-high-risk">High Risk | Measured                                      | Rating                                          |
|----|----------------------------|---------------------------------|---------------------------------------|-----------------------------------------------|-------------------------------------------------|-----------------------------------------------|-------------------------------------------------|
| H1 | High Cyclomatic Complexity | Max complexity per method       | <10                                   | 10–20                                         | >20                                             | ~12 (ConnectPage.tsx)                         | <span class="rating rating-moderate">Moderate   |
| H2 | Large Classes              | Largest class/file LOC          | <300                                  | 300–1000                                      | >1000                                           | 276 LOC (dev-api/src/server.js)               | <span class="rating rating-good">Good           |
| H3 | Large Functions            | Largest function/method LOC     | <50                                   | 50–200                                        | >200                                            | 214 LOC (ConnectPage)                         | <span class="rating rating-high-risk">High Risk |
| H4 | Business Logic Duplication | Duplicated business logic %     | <5%                                   | 5–10%                                         | >10%                                            | ~1.3% (buildTree ×3, reachability calc ×3)    | <span class="rating rating-good">Good           |
| H5 | Duplicate Code (general)   | Overall duplicate code %        | <5%                                   | 5–10%                                         | >10%                                            | ~5% (structural + copy-paste)                 | <span class="rating rating-moderate">Moderate   |
| H6 | High Churn Areas           | Monthly changes (top files)     | <5                                    | 5–10                                          | >10                                             | 2 (server.js, api.php, ConnectController.php) | <span class="rating rating-good">Good           |
| H7 | Defect-Prone Files         | Fix commits (hottest file)      | 1–3                                   | 4–5                                           | >5                                              | 1 (App.tsx)                                   | <span class="rating rating-good">Good           |
| H8 | Ownership Issues           | Top-author ownership %          | >80%                                  | 60–80%                                        | <60%                                            | 100% (single author: ksabai-gl)               | <span class="rating rating-good">Good           |
| H9 | Resource Leak (additional) | Components with missing cleanup | 0                                     | 1                                             | >1                                              | 1 (LegacyMonitorPoller.jsx)                   | <span class="rating rating-moderate">Moderate   |

#### Hotspot Score breakdown

| Component             | Weight | Sub-score (0–100) | Weighted |
|-----------------------|--------|-------------------|----------|
| Cyclomatic Complexity | 25%    | 40                | 10.0     |
| Code Churn            | 25%    | 10                | 2.5      |
| Defect Density        | 20%    | 15                | 3.0      |
| Class/Function Size   | 15%    | 70                | 10.5     |

|                            |      |    |          |
|----------------------------|------|----|----------|
| Business Logic Duplication | 10%  | 40 | 4.0      |
| Developer Ownership Risk   | 5%   | 5  | 0.25     |
| Hotspot Score              | 100% |    | 30 / 100 |

2.5 Actions Required

| Hotspot                         | Action                                                                                                                                 | Rating                                          | Priority                                |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|-----------------------------------------|
| H3 — Large Functions            | Decompose ConnectPage (214 LOC) and DiscoveryPage (166 LOC) into sub-components and custom hooks, each under 80 LOC                    | <span class="rating rating-high-risk">High Risk | <span class="sev sev-critical">Critical |
| H1 — High Cyclomatic Complexity | Extract conditional rendering sections into guard components; flatten nested ternaries with early returns                              | <span class="rating rating-moderate">Moderate   | <span class="sev sev-medium">Medium     |
| H5 — Duplicate Code             | Create shared page-layout composition and extract common query/mutation hooks; remove or migrate LegacyDashboardWidget                 | <span class="rating rating-moderate">Moderate   | <span class="sev sev-medium">Medium     |
| H9 — Resource Leak              | Add componentWillUnmount cleanup to LegacyMonitorPoller or convert to function component; add Error Boundary for LegacyDashboardWidget | <span class="rating rating-moderate">Moderate   | <span class="sev sev-medium">Medium     |
| H4 — Business Logic Duplication | Consolidate reachability calculation into ConnectMonitor model method; extract buildTree into shared service or trait                  | <span class="rating rating-good">Good           | <span class="sev sev-low">Low           |

2.6 Expected Outcomes

- **Lower defect risk on page changes:** Decomposing ConnectPage and DiscoveryPage into focused sub-components reduces the number of test paths per unit from ~12 to ~3–4, cutting regression risk on UI changes.
- **Single-source business rules:** Consolidating reachability calculation and tree-building logic eliminates the risk of formula drift across three codebases (Laravel, dev-API, controller).
- **Safer reviews:** Smaller, focused components and hooks are easier to review in PRs — reviewers can assess one concern at a time rather than parsing a 214-line monolith.
- **Eliminated resource leaks:** Fixing the interval leak in LegacyMonitorPoller prevents accumulated network traffic and React warnings in development, improving runtime stability.
- **Clearer architecture for growth:** Shared layout patterns and extracted hooks establish a composition model that scales cleanly as Discovery and Connect modules add features (filtering, export, bulk operations).  
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